

CHANGLING, China

WMO: 540490

Lat: 44.2784N

Lon: 123.9671E

Elev: 623

StdP: 14.37

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-9.3	-23.9	1.5	-7.1	-20.3	1.8	-4.2	20.5	12.0	18.1	15.7	3.4	250	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	89.3	69.5	86.7	69.2	84.5	68.8	76.1	82.9	74.6	81.2	73.1	79.6	8.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	130.0	80.6	72.4	122.8	79.1	70.9	116.6	77.7	40.2	83.0	38.7	81.3	37.3	79.6	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
20.1	17.3	15.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-16.9	93.4	5.6	3.2	-20.9	95.7	-24.2	97.6	-27.4	99.4	-31.5	101.7
			-17.3	77.9	5.5	1.9	-21.3	79.2	-24.5	80.3	-27.6	81.4	-31.6	82.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	5.9	14.9	29.9	47.5	61.1	71.4	75.5	72.4	62.1	46.5	26.2	9.9	
	DBStd	25.76	8.53	9.76	10.06	8.98	7.41	5.72	4.16	4.76	6.70	8.91	10.98	8.89	
	HDD50	5265	1366	982	626	152	5	0	0	0	4	173	714	1243	
	HDD65	8610	1831	1402	1088	527	167	11	1	6	130	576	1163	1708	
	CDD50	2986	0	0	2	77	349	642	790	694	367	64	1	0	
	CDD65	858	0	0	0	2	46	204	326	235	44	1	0	0	
	CDH74	6844	0	0	0	32	467	1774	2619	1644	298	10	0	0	
	CDH80	1926	0	0	0	3	120	614	772	381	36	0	0	0	
Wind	WSAvg	6.8	5.5	6.4	7.9	9.0	8.5	6.8	6.0	5.5	6.0	6.9	6.8	5.8	
Precipitation	PrecAvg	18.0	0.0	0.1	0.3	0.7	1.5	2.7	6.2	3.7	1.7	0.7	0.2	0.1	
	PrecMax	28.2	0.2	0.5	1.1	4.8	4.0	4.5	11.5	8.5	4.8	2.5	0.8	0.5	
	PrecMin	10.9	0.0	0.0	0.0	0.0	0.2	0.7	2.7	0.7	0.2	0.0	0.0	0.0	
	PrecStd	4.2	0.0	0.2	0.4	0.9	0.9	1.1	2.4	2.2	0.9	0.6	0.2	0.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	44.7	63.4	78.1	87.5	94.4	92.2	89.3	83.9	75.0	57.5	39.4	
		MCWB	27.5	33.9	46.5	54.3	61.5	66.6	73.3	73.2	66.4	58.5	47.4	35.9	
	2%	DB	26.4	39.2	55.1	72.3	83.6	89.9	89.1	86.2	80.2	68.6	52.0	31.9	
		MCWB	22.0	30.7	41.0	51.4	59.2	65.5	72.5	71.5	64.7	55.5	43.2	28.0	
	5%	DB	22.6	34.9	50.2	67.8	79.4	86.6	86.7	84.0	77.4	64.2	46.8	27.5	
		MCWB	18.8	28.0	38.1	49.7	58.1	65.3	72.1	71.0	62.8	52.3	40.0	23.5	
	10%	DB	18.9	30.7	45.7	63.5	75.2	83.5	84.4	81.9	74.6	60.4	41.9	23.1	
		MCWB	15.8	25.2	35.1	47.5	55.8	64.9	71.7	70.0	61.0	50.5	35.7	19.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.5	35.4	47.6	57.8	65.7	72.5	78.9	77.7	71.4	61.6	49.7	35.8
			MCDB	31.6	42.1	59.6	70.3	80.9	83.6	87.3	84.2	79.4	68.7	55.1	38.4
2%		WB	22.2	31.5	42.5	54.2	62.5	70.5	76.8	75.8	68.6	57.9	45.0	28.5	
		MCDB	25.9	38.3	53.0	68.2	76.5	81.4	83.6	82.2	76.0	65.9	49.7	31.6	
5%		WB	18.9	28.3	38.8	51.2	60.2	69.0	75.3	74.3	66.3	55.2	40.5	23.6	
		MCDB	22.4	34.8	49.2	65.1	74.2	79.4	82.3	80.3	72.9	61.9	46.2	27.4	
10%		WB	15.8	25.0	35.7	48.3	58.1	67.6	73.9	72.8	63.8	52.0	35.9	19.7	
		MCDB	18.9	30.7	45.1	61.5	71.8	77.7	80.7	78.6	71.2	59.0	41.5	23.0	
Mean Daily Temperature Range	5% DB	MDBR	17.3	18.6	18.8	20.5	19.8	17.3	14.5	15.6	19.7	18.6	16.2	16.0	
		MCDBR	19.6	21.3	23.9	25.9	24.9	22.7	17.9	17.9	21.8	22.3	19.6	18.4	
		MCWBR	16.4	15.7	14.3	13.1	10.2	7.9	6.5	7.4	9.5	12.1	13.6	15.4	
	5% WB	MCDBR	19.2	20.8	22.3	22.2	20.6	16.4	13.5	13.7	17.0	19.0	18.9	18.3	
		MCWBR	16.3	15.7	13.9	12.7	10.1	6.9	6.6	7.3	9.5	12.0	14.3	15.7	
Clear-Sky Solar Irradiance	taub	0.310	0.339	0.410	0.511	0.587	0.651	0.605	0.537	0.471	0.453	0.369	0.323		
	taud	2.138	2.095	1.960	1.772	1.666	1.625	1.765	1.910	1.985	1.920	2.056	2.095		
	Ebn at Noon	248	262	256	239	225	212	220	231	235	215	219	226		
	Edh at Noon	30	38	50	67	76	80	69	57	49	46	33	29		
All-Sky Solar Radiation	RadAvg	700	1025	1388	1686	1870	1797	1708	1604	1427	1022	685	573		
	RadStd	62	73	71	128	128	150	145	108	100	64	56	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	-1.93	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page